

Suggested Biomedical Protocol for Autism Spectrum Disorder

A Biomedical, Dietary & Therapeutic Protocol — Model for Clinical Reference

*"We cannot undo the original exposure, but we can reverse the
cascade at multiple intervention points simultaneously."*

— Suggested Protocol for Clinical Reference —

Suggested Use Case: Male, 7 years old, Level 3 Autism, Non-Verbal, ~25 kg

Based on: Deep analysis of 301 research papers

Date: May 31, 2026

Version: v1.5 — Shareable Protocol Model

Note: Doses and schedules shown are for the suggested use case. See Appendix for weight-based scaling.

Suggested Biomedical Protocol v1.5 — White Paper

This document is for informational purposes only. Not medical advice. All treatments should be discussed with qualified medical professionals before implementation.

About this document: This is a suggested treatment protocol based on analysis of 301 research papers. The patient profile (7yr male, Level 3, ~25kg) is an illustrative use case. All dosing in the appendix is expressed per kilogram for adaptation to individual patients. **This is not medical advice.** All treatments require qualified medical supervision.

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Part 1: The Framework

1.1 Our Working Hypothesis

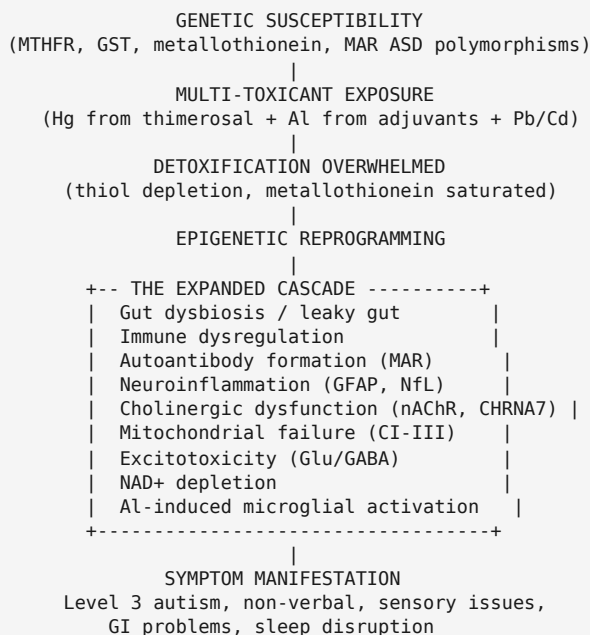
Based on cumulative evidence from 301 research papers (291 original + 10 on nicotinic/cholinergic pathways):

Your son's autism is driven by multi-toxicant environmental exposure — primarily mercury from thimerosal plus aluminum from vaccine adjuvants — that exceeded his natural detoxification capacity during critical neurodevelopmental windows. This was amplified by genetic susceptibility, gut microbiome disruption, and resulting epigenetic reprogramming that locked in the pathological state.

This does NOT mean this was anyone's fault. It means a specific combination of genetic susceptibility plus environmental exposure produced a biological cascade that his body could not handle.

The good news: Treatment targets this cascade at multiple points simultaneously, and epigenetic changes are potentially reversible with intensive biomedical intervention.

1.2 The Expanded Cascade Model



1.3 The Treatment Logic

#	Intervention	Cascade Target
1	Remove ongoing toxicant exposure	Source
2	Methylation + glutathione support	Detoxification
3	Speciation-guided chelation (DMSA, ALA, silica)	Stored heavy metals
4	GFCF diet + gut healing → FMT if needed	Gut dysbiosis
5	Anti-inflammatory agents (LDN, curcumin)	Neuroinflammation
6	Sulforaphane, taurine, low-glutamate diet	Excitotoxicity
7	MCT oil, CoQ10, NAD+ precursors	Mitochondria
8	IVIG, LDN, LDI (if autoantibodies+)	Autoimmunity
9	ABA, speech, OT (synergy-timed)	Development
10	Melatonin, magnesium, vagal therapy	Symptoms

1.4 Guiding Principles

1. **First, do no harm** — start with safest interventions, progress slowly
2. **Test before treating** — biomarkers guide every decision
3. **Synergy over isolation** — biomedical + behavioral together work best
4. **Sequence matters** — detox first, then development follows
5. **Individualize** — every child's biochemical profile differs
6. **Epigenetics are malleable** — methylation support can reverse
7. **The gut is the foundation** — heal it before deeper treatments

Part 2: The Three-Phase Protocol

2.1 Phase 1 — Foundation (Months 1-3)

Goal: Stabilize the child's body, establish baseline testing, prepare for deeper intervention.

2.1A Dietary Protocol

Week	Intervention	Details
1-2	Remove dairy	All milk, cheese, yogurt, butter. A2 milk or goat milk trial after 6 weeks
3-4	Remove gluten	All wheat, barley, rye. Replace with rice, quinoa, buckwheat
4+	Remove processed	Artificial colours, preservatives, MSG, HFCS
1+	Bone broth	1 cup daily — gut healing (glycine, glutamine)
2+	MCT oil	Start ½ tsp/day → 2 tsp 2x/day
1+	Epsom baths	2 cups in warm water, 20 min, 4-5x/week

2.1B Initial Supplements

Supplement	Daily Dose	Timing	Purpose
Vitamin D3	3,000 IU	Morning	Immune, neurodevelopment
Omega-3 (DHA)	750 mg	Breakfast	Brain, anti-inflammatory
Magnesium glycinate	200 mg	Bedtime	Sleep, NMDA blockade
Zinc picolinate	15 mg	Dinner	Metallothionein, immune
Probiotic (multi-strain)	50 bil CFU	Between meals	Microbiome restoration
Digestive enzymes	1 capsule	With meals	Nutrient absorption
NAC	250 mg 2x	Detox	Glutathione precursor
Niacinamide	50 mg	Breakfast	NAD+ precursor
Selenium	50 mcg	Breakfast	GPX cofactor
MCT oil	1 tsp 2x→1 tbsp	With food	Mitochondrial fuel

2.1C Baseline Testing

Test	Measures	Lab / Code	Priority
Urinary porphyrins	Mercury toxicity	Doctor's Data / Great Plains	Essential
Provoked urine metals	Hg, Pb, Cd, Al	Doctor's Data	Essential
Hair mineral analysis	Minerals + toxic metals	Trace Elements / Great Plains	Essential
Thiol/disulfide ratio	Detox capacity	Doctor's Data	Essential
MTHFR C677T + A1298C	Methylation genetics	23andMe / SpectrumDNA	Essential
Vitamin D, Zn, Cu, ferritin	Nutritional status	LabCorp / Quest	Essential
Comprehensive stool (CDSA)	Dysbiosis, Candida	Doctor's Data / Genova	Essential
Food Sensitivity (IgG) Panel	Reactive foods	Great Plains / US Biotech	Recommended
Thyroid Panel	TSH, free T3, free T4	LabCorp / Quest	Recommended
GFAP + NfL	Neuroinflammation	Specialty lab	Recommended
Plasma glutamate	Excitotoxicity	Specialty lab	Recommended
Cu:Zn ratio	Metallothionein	LabCorp / Quest	Recommended
MAR ASD panel	Autoantibodies	Great Plains	If indicated
tTG-2 IgA	Autoimmune gluten	LabCorp / Quest	Recommended
Isoprostane iPF2α-VI	Oxidative stress	Doctor's Data	Recommended
Serum/urine aluminum	Al body burden	Doctor's Data	If indicated

2.1D Therapy Setup

Therapy	Frequency	Focus
ABA	20-30 hrs/week	AAC, self-help, behavior
Speech	3x/week	PECS → AAC system
Occupational therapy	2x/week	Sensory, feeding
PECS/AAC	Daily — start now	Functional communication
Vagal therapy	Daily	Gargling, cold water, humming

Daily tracking baseline (start immediately):

Metric	How	Frequency
ATEC score	Free online form	Monthly — decreasing = progress
Functional communications	Intentional acts: PECS, AAC, sounds	Weekly — target 50+/day
Eye contact events	Spontaneous, sustained	Daily
Stimming episodes	Duration + frequency	Weekly — should decrease
Self-injurious / aggressive	Incident count	Daily — trend to zero
Sleep diary	Hours, night wakings, onset	Daily — target 8-10 hrs
Bowel diary (Bristol type)	Type 1-7 + frequency	Daily — target 3-4, 1-2x/day
Diet compliance	% GFCF meals	Daily — 90%+ target

Bristol Stool Chart reference:

Type	Description	Meaning
1-2	Hard lumps / sausage	Constipation — increase Mg, water
3-4	Smooth, soft / snake	Ideal — healthy transit
5	Soft blobs, clear edges	Normal / slightly fast
6	Mushy, fluffy pieces	Fast transit — possible issue
7	Watery, no solids	Diarrhea — reduce dose, check

2.1E Sleep Protocol

Intervention	Purpose
Melatonin 1-10 mg at bedtime	Sleep initiation + neuroprotection
Magnesium glycinate 200 mg	Calming, sleep quality
Strict bedtime routine (visual schedule)	Predictability
Dark room, no screens 1 hr before bed	Circadian support
Weighted blanket	Sensory calming

2.2 Phase 2 — Active Treatment (Months 4-12)

Goal: Remove accumulated toxicants, correct neurochemistry, intensive therapy during biomedical windows.

2.2A Methylation Support

Agent	Dose	Mechanism
Methyl B12 injection	1,000 mcg every 3 days	Direct methyl donor
Sublingual B12 (alt.)	5,000 mcg daily	Alternative route
L-methylfolate	400-800 mcg daily	Bypasses MTHFR block
TMG	500-1,000 mg daily	Alternative donor
SAMe	200-400 mg daily	Direct donor

2.2B Glutathione & Thiol Restoration

Agent	Dose	Role
Liposomal glutathione	500 mg/day	Direct replacement
OR Nebulized glutathione	100 mg 1-2x/day	Lung/liver detox
NAC	500 mg 2x/day	Thiol precursor
Alpha-lipoic acid	100 mg/day	Glutathione recycling, brain Hg binder
Selenium	50 mcg/day	GPX cofactor

Monitor: Thiol/disulfide ratio every 3 months (target: normalized).

2.2C NAD+ Support

Agent	Dose	Mechanism
Nicotinamide riboside (NR)	125 mg/day	NAD+ via NRK pathway
Apigenin	50 mg 2x/day	PARP1 inhibitor — preserves NAD+
Niacin (optional)	25 mg 2x/day	NAD+ via Preiss-Handler

2.2D Mitochondrial Support

Agent	Dose	Target
CoQ10 (ubiquinol)	100-200 mg/day	Complex I, III support
L-carnitine	500 mg 2x/day	Fatty acid transport
Creatine monohydrate	2-3 g/day	ATP reserve
B-complex	1 cap/day	Cofactors
MCT oil	1-2 tbsp/day	Ketones bypass Complex I

2.2E Anti-Excitotoxicity Protocol

Agent	Dose	Mechanism
Sulforaphane	10-50 mg/day	Upregulates glutamate transporters
Taurine	250-500 mg 2x/day	GABA-A agonist
Low-glutamate diet	Avoid MSG, hydrolyzed proteins	Reduces excitotoxic load
Magnesium glycinate	200-400 mg bedtime	NMDA blockade
Memantine (off-label)	2.5-10 mg/day	NMDA antagonist (specialist only)

2.2F Gentle Detox (Months 4-6)

Agent	Dose	Target
Chlorella	2 g 2x/day (titrate from 500 mg)	Metal binding
Cilantro tincture	10-15 drops 2x/day	Hg mobilization
Silica-rich water	1 L/day (Fiji, Spritzer)	Aluminum chelation
Malic acid	500 mg 2x/day with meals	Al + Krebs cycle
Epsom baths	2 cups, 20 min, 5x/week	Sulfate support
Infrared sauna	10-20 min, 3-5x/week	Sweat elimination

2.2G Heavy Metal Chelation (Month 6+)

Agent	Dose	Schedule	Target
DMSA	10 mg/kg 3x/day	3 on / 11 off	Hg, Pb, Cd
ALA	100 mg/day	Same pulse	Brain Hg
Silica + malic acid	Per protocol	Off-days	Aluminum

Note: Only under DAN! or MAPS physician. 8-12 rounds minimum. Replace minerals on off-days (Zn, Se, Cu). Monitor liver/kidney monthly. **Schedule intensive ABA on pulse days 2-3** for brightening window synergy.

2.2H GI Treatment

Condition	Treatment	Duration
Candida overgrowth	Nystatin or Diflucan	2-4 weeks
SIBO	Rifaximin 550 mg 3x/day	14 days
SIFO	Antifungal + SCD diet	4-8 weeks
Refractory GI (6 mo+)	FMT referral	1 procedure
Gut lining repair	L-glutamine, Zn carnosine	Ongoing

Yeast die-off management (Herxheimer reaction):

Symptom	Intervention
Worsening behavior, irritability	Activated charcoal 250 mg between meals, increased water
Severe irritation	Reduce antifungal 50%, restart full dose after 3 days
Lethargy	Increase B12 temporarily, ensure electrolytes
Constipation	Increase magnesium glycinate dose

2.2I Immune Modulation

Autoantibody Status	Protocol
MAR ASD + or brain ab +	IVIG 400 mg-2 g/kg/month
Mild autoimmunity only	LDN 1.5-4.5 mg at bedtime + LDI
No autoantibodies	Anti-inflammatory support only

Anti-Inflammatory Support (add Month 6+ if not started):

Agent	Dose	Mechanism	Notes
Luteolin	100 mg 2x/day	Mast cell stabilizer, brain anti-inflammatory	Natural flavonoid (celery, green pepper)
Quercetin	250-500 mg 2x/day	Mast cell stabilizer, anti-inflammatory	Often combined with luteolin
Curcumin	500-1,000 mg/day	NF-κB inhibition	With piperine for absorption. Avoid if gallstones

2.2J Therapy Synergy

Window	Maximum	Rationale
DMSA pulse days 2-3	ABA 30-40 hrs/week	Peak cognitive brightening
24-72 hrs post Methyl B12	Speech therapy	Methylation window
Epsom bath evenings	Vagal therapy, AAC	Post-detox calm
Daily	AAC 50+ comms	Communication foundation

2.3 Phase 3 — Maintenance (Months 12-24+)

Goal: Sustain gains, continue development, prevent regression.

Treatment	Frequency
DMSA pulses	1-2 rounds every 3 months
Methyl B12	Weekly maintenance
GFCF diet	Continue or A2 reintroduction trial
Supplements (vit D, omega-3, Mg, NAD+, MCT)	Maintain
Antifungal pulses	As needed
Sulforaphane maintenance	Ongoing
Advanced: IVIG, HBOT, FMT	If indicated
School transition	Integrated or special education
AAC → spoken language bridge	Ongoing

2.4 Advanced Adjunctive Options (Specialist Supervision Required)

The following treatments are supported by emerging research but require specialist oversight. Do NOT attempt without a qualified physician.

2.4A Cholinergic/Nicotinic nAChR Modulation

Rationale: Nicotinic acetylcholine receptors (nAChRs) are reduced in ASD — postmortem studies show 40-70% lower α7 nAChR binding in cortex and thalamus. The *CHRNA7* gene (15q13.3) is a known ASD susceptibility locus. Heavy metals (Hg, Cd, Pb) directly bind and inhibit nAChRs, creating cholinergic dysfunction that may underlie cognitive, attentional, and sensory-processing deficits.

How existing treatments support nAChR recovery:

Existing Protocol Component	nAChR Benefit
Methylation support (B12, folate)	<i>CHRNA7</i> gene expression requires methylation
Glutathione restoration	nAChR function is redox-sensitive — requires GSH
Heavy metal chelation	Removing Hg/Cd/Pb unblocks inhibited nAChRs
Anti-excitotoxicity (sulforaphane, taurine)	nAChRs regulate glutamate/GABA balance
NAD ⁺ support (NR, apigenin)	nAChR signaling depends on NAD ⁺ /sirtuin pathways

Transdermal nicotine — experimental option (Month 12+, MAPS/DAN! physician only):

Parameter	Detail
Agent	Transdermal nicotine patch (NOT smoking)
Starting dose	3.5 mg/day (obtain child-appropriate dosing from pharmacy)
Titration	Increase by 3.5 mg every 2 weeks, max 14 mg/day
Duration	12-week trial, reassess for benefit
Target symptoms	Aggression, irritability, attention, sensory processing
Evidence	Lewis 2018 (n=40, reduced irritability vs placebo); Van Schalkwyk 2015 (adolescent pilot)
Safest alternative	Cotinine 2-4 mg/day (Berger 2021 — avoids rewarding properties)
Monitoring	Heart rate, blood pressure weekly; behavior scales monthly

Risks & contraindications: Nicotine is addictive. Not suitable for children under 12 (no safety data). Contraindicated in cardiovascular disease, asthma, pregnancy. Requires documented aggression/irritability not responding to other treatments. Side effects: nausea, dizziness, headache, sleep disturbance, skin irritation. *For most children, the existing protocol (methylation + glutathione + chelation + NAD⁺) may gradually restore nAChR function without direct nicotine intervention.*

Part 3: Expectations & Outcomes

3.1 Realistic Timeline

Timeframe	Reliable Goal	Stretch Goal
0-3 months	Better sleep, less GI distress, more eye contact	5-10 PECS requests/day
3-6 months	Reduced stimming, AAC begins	20+ comms/day
6-12 months	Functional AAC, calm, follows routines	First words
12-18 months	Toilet training, social interest	2-word phrase (AAC)
18-24 months	Self-help, longer attention	Simple conversation (AAC)

3.2 Factors Predicting Better Response

Biomarker	Predicts
GFAP decline at 3 months	Strong anti-inflammatory response
Thiol/disulfide normalization	Strong chelation response
Cu:Zn ratio correction	Cognitive/behavioral improvement
GI improvement on GFCF	FMT benefit if escalated
Brightening during DMSA	Overall good chelation outcome

3.3 Weekly Schedule (DMSA Pulse Week)

Time	Mon (Pulse D1)	Tue (Pulse D2)	Wed (Pulse D3)	Thu-Fri	Sat-Sun
8 AM	DMSA + Morning	DMSA + Morning	DMSA + Morning	Standard	Standard
9-12	ABA (peak)	ABA + speech	ABA (peak)	ABA continues	OT / nature
12-1	Lunch	Lunch	Lunch	Lunch	Lunch
1-3	OT + AAC	Speech (max)	Floortime + AAC	Therapy	Family time
3:30	Snack	Snack	Snack	Snack	Snack
4-5:30	Vagal therapy	Sensory/park	Music therapy	Free play	Park
5:30-6	Dinner	Dinner	Dinner	Dinner	Dinner
6-7	Epsom bath	Epsom bath	Epsom bath	Epsom bath	Epsom bath
7-8	Cranial hold	AAC + wind down	Cranial hold	Wind down	Wind down
8 PM	Melatonin + bed	Melatonin + bed	Melatonin + bed	Melatonin + bed	Melatonin + bed

Weekly totals: ABA 25-30h, Speech 3h, OT 3h, Epsom 5-6x, MCT daily, Silica water daily

Part 4: Testing & Biomarker Tracking

Biomarker	What It Tells Us	Frequency	Target
ATEC score	Overall severity	Monthly	Decreasing
Communication count	AAC/speech progress	Weekly	50+ acts/day
Sleep diary	Hours, quality	Daily	8-10 hrs
Bowel diary	Frequency, consistency	Daily	1-2 formed stools
Porphyrins	Hg toxicity	6 months	Normalized
Provoked urine metals	Hg, Pb, Cd, Al	6-12 months	Decreasing
Thiol/disulfide	Hg detox capacity	3 months	Normalized
Isoprostane (iPF2 α -VI)	Oxidative stress	3 months	Decreasing
tTG-2 IgA	Autoimmune gluten	6 months	Negative
GFAP + NfL	Neuroinflammation	6 months	Decreasing
Plasma glutamate	Excitotoxicity	6 months	<50 μ mol/L
Cu:Zn ratio	Metallothionein	3 months	0.8-1.2
Quinolinic/kynurenic acid	Kynurenine pathway	6 months	Normalized
8-OHdG	DNA oxidative damage	3 months	Decreasing

Part 5: The Pulsed-Bolus Hypothesis

5.1 Multi-Toxicant Pulsed-Bolus Model

The routine vaccine schedule creates pulsed-bolus exposure of BOTH mercury AND aluminum:

Age	Vaccines	Hg (μ g)	Al (mg)
Birth	Hep B	12.5	0.25
2 mo	DTaP, IPV, Hib, PCV, RV	62.5	1.2
4 mo	DTaP, IPV, Hib, PCV, RV	50.0	1.2
6 mo	DTaP, IPV, Hib, PCV, RV	62.5	1.2
12 mo	MMR (Hg-free), PCV, Hep A	0	0.5
15 mo	DTaP, IPV, Hib	37.5	0.85
18 mo	Booster	12.5	0.25
Total		187.5 μg	5.5 mg

Each bolus arrives before the previous one is cleared. In a susceptible child: mercury accumulates (brain half-life years), aluminum persists at injection sites (MMF lesions). Combined neuroinflammatory burden exceeds threshold → regression.

5.2 Treatment Corollaries

1. **DMSA chelation** addresses Hg + Pb + Cd body burden
2. **Silica + malic acid** addresses aluminum specifically

3. **Methylation + glutathione** restores endogenous detox capacity
4. **NAD+ support** enables DNA repair and epigenetic reversal
5. **Anti-excitotoxicity** corrects glutamate/GABA imbalance
6. **Therapy synergy** maximizes gains during brightening windows

Appendix: Weight-Based Dosing Reference

For adapting this plan to children of different ages and weights. Shareable with other families.

A1: Supplement Dosing by Weight

Supplement	Per kg/day	10-15 kg	15-20 kg	20-30 kg *	30-45 kg	Safety
Vitamin D3	100-150 IU	1,000-2,000 IU	1,500-3,000 IU	2,000-4,000 IU	3,000-6,000 IU	Max 4k IU <8yr
Omega-3 DHA	20-30 mg	200-300 mg	300-500 mg	500-800 mg	800-1,200 mg	Anticoagulant caution
Mg glycinate	8-15 mg	100-200 mg	150-300 mg	200-400 mg	300-600 mg	Reduce if loose stools
Zn picolinate	0.5-0.8 mg	5-10 mg	10-15 mg	15-25 mg	20-30 mg	Max 40 mg/day
NAC	10-20 mg	100-200 mg	200-300 mg	250-500 mg	400-800 mg	GI upset possible
L-methylfolate	15-30 mcg	150-400 mcg	300-600 mcg	400-800 mcg	600-1,200 mcg	Max 1,000 mcg
Selenium	2-4 mcg	20-40 mcg	35-70 mcg	50-100 mcg	70-140 mcg	Toxic >400 mcg
CoQ10 ubiquinol	4-8 mg	40-80 mg	60-120 mg	100-200 mg	150-350 mg	Take with food
Melatonin	0.1-0.5 mg	1-5 mg	1.5-7 mg	2-10 mg	3-15 mg	Start low
Taurine	15-30 mg	150-300 mg	200-500 mg	250-500 mg	400-800 mg	Very safe
Niacinamide	2-4 mg	20-50 mg	30-70 mg	50-100 mg	70-150 mg	No flush
NR (nicotinamide riboside)	3-6 mg	30-60 mg	50-100 mg	75-150 mg	100-200 mg	Expensive

★ 20-30 kg = target range for this plan (7-year-old)

A2: Chelation Agent Dosing

Agent	Per kg	Freq	10-15 kg	15-20 kg	20-30 kg	30-45 kg
DMSA	10 mg x3/day	Every 8h	100-150 x3	150-200 x3	200-300 x3	300-450 x3
ALA	3-5 mg/day	Daily	30-50 mg	45-75 mg	60-100 mg	90-150 mg
Malic acid	20-30 mg/day	2x/day	200-300 mg	300-450 mg	500-700 mg	700-1,000 mg

Cycle: DMSA + ALA = 3 days on / 11 days off. Malic acid daily for aluminum protocol.

A3: Pharmaceutical Interventions

Agent	Per kg	Typical (20-30 kg)	Notes
LDN	0.05-0.2 mg	1.5-4.5 mg at bedtime	Compounding pharmacy
Sulforaphane	0.5-2 mg/day	10-50 mg/day	Standardized extract
Nystatin susp.	100k-200k U/day	2-4 mL 4x/day	GI only
Diflucan	3-6 mg/day	50-150 mg/day	Liver monitoring
Memantine	0.1-0.4 mg/day	2.5-10 mg/day	Specialist only
IVIG	400 mg-2 g/month	10-50 g/month	Very expensive
Methyl B12 inj.	25-50 mcg	500-1,000 mcg	Every 3 days

A4: MCT Oil Titration

Weight	Starting	Target	Titration
10-15 kg	¼ tsp 1x/day	1-2 tsp/day	+¼ tsp every 5-7 days
15-20 kg	½ tsp 1x/day	1-2 tsp/day	+¼ tsp every 5-7 days
20-30 kg	½ tsp 2x/day	1-2 tbsp/day	+½ tsp every 5-7 days
30-45 kg	1 tsp 2x/day	1-3 tbsp/day	+1 tsp every 5-7 days

Side effect: Loose stools — reduce if problematic, titrate slower. Mix into food.

A5: Silica Water for Aluminum Detox

Weight	Daily Volume	Sources
10-20 kg	500-750 mL	Fiji, Spritzer, Volvic
20-30 kg	750-1,000 mL	Divide with meals
30-45 kg	1,000-1,500 mL	Throughout day

Mechanism: Orthosilicic acid binds aluminum → excreted renally.

A6: Epsom Salt Baths by Weight

Weight	Salt	Temp	Duration	Frequency
10-15 kg	1 cup	35-37°C	10-15 min	3-5x/week
15-20 kg	1½ cups	35-37°C	15-20 min	4-5x/week
20-30 kg	2 cups	35-37°C	15-20 min	5-6x/week
30-45 kg	2-3 cups	35-37°C	20-30 min	5-6x/week

A7: Mineral Replacement (Chelation Off-Days)

Mineral	Per kg/off-day	Form
Zinc	1 mg/kg/day	Picolinate
Copper	0.04-0.08 mg/kg/day	Glycinate
Selenium	2-4 mcg/kg/day	Selenomethionine
Molybdenum	2-4 mcg/kg/day	Glycinate

A8: Age-to-Weight Reference

Age	Avg Weight	Plan Target Range
1 year	10 kg	—
3 years	14-15 kg	10-15 kg
5 years	18-19 kg	15-20 kg
7 years	22-25 kg	20-30 kg
9 years	28-32 kg	30-45 kg
12 years	40-50 kg	Adult dosing

A9: Emergency Symptoms — Stop Treatment & Seek Medical Care

- **Allergic reaction:** Hives, facial swelling, difficulty breathing
- **Severe lethargy:** Difficult to arouse
- **Decreased urination:** Less than 3 voids in 24 hours
- **Persistent vomiting:** More than 3 episodes in 12 hours
- **Seizure:** New onset or increased frequency
- **Blood:** In stool or urine

A10: Side Effects Reference — What to Expect Week by Week

This section details known and potential side effects for each treatment, organized by treatment category. Not every child experiences these — many tolerate treatments well. The goal is preparation, not alarm.

A10.1 Dietary Changes (Weeks 1-4)

Intervention	Expected	Common (30-60%)	Occasional (10-30%)	Rare (<10%)	Management
Dairy removal	None	Increased tantrums (days 2-5), craving for restricted foods	Transient constipation, asking for old foods repeatedly	Weight loss >5%	Offer dairy-free alternatives, distract with preferred activities, ensure caloric intake
Gluten removal	None	Irritability (days 3-7), mood swings	Fatigue, headaches, brain fog	Severe withdrawal-like behavior	Taper gluten over 1 week instead of cold turkey, offer GF alternatives
GFCF combined	None	Picky eating intensifies before improving (days 5-14)	Constipation, reduced appetite	Refusal to eat (rare, monitor)	Offer safe foods alongside new ones, bone broth for nutrition, patience
MCT oil introduction	Loose stools if starting too fast	Soft stools (first 3-5 days), flatulence, bloating	Nausea, abdominal cramping	Diarrhea requiring dose reduction	Start ¼ tsp, increase by ¼ tsp every 5-7 days. Reduce if stools become watery. Mix into applesauce or warm food
Bone broth daily	None	None typically	Mild increase in bowel movements	Allergic reaction (very rare — fish/shellfish broth)	Use chicken or beef broth if fish allergy suspected

Expected timeline for dietary adjustment: Most children adapt within 2 weeks. The first 3-5 days are hardest. After 14 days, cravings subside and the new diet becomes routine.

A10.2 Phase 1 Supplements (Months 1-3)

Supplement	Expected	Common (30-60%)	Occasional (10-30%)	Rare (<10%)	Management
Vitamin D3 3,000 IU	None	None	Mild nausea if on empty stomach	Toxicity (months of overdose only)	Give with food. Blood test at 3 months.
Omega-3 DHA 750 mg	None	Fishy burps, fishy breath	Loose stools, nausea	Allergic reaction (fish allergy)	Freeze capsules, give with food, switch to algal DHA if fish-sensitive
Magnesium glycinate 200 mg	Calming effect	Softer stools (within 3-5 days)	Drowsiness (daytime if dose too high)	Diarrhea, lethargy	Reduce dose if stools watery. Move to bedtime only if daytime drowsy
Zinc picolinate 15 mg	None	None	Nausea (on empty stomach), metallic taste	Vomiting	Always give with dinner. If nausea persists, reduce to 10 mg
Probiotic 50 bil CFU	None	Mild bloating (days 1-3), increased gas	Transient loose stools, mild abdominal discomfort	None significant	Start with 25 billion for first week. If severe bloating persists, switch brands
Digestive enzymes	None	None typically	Mild stomach upset	Allergic reaction (rare — enzyme source allergy)	Give with first bite of meal, not before
NAC 250 mg 2x/day	None	None	Nausea, loose stools (rare)	Skin rash (very rare)	Give with food. Reduce to 1x/day if GI upset. Stop if rash
Niacinamide 50 mg	None	None (no flush like niacin)	Mild nausea on empty stomach	None significant	Give with breakfast. Niacinamide does NOT cause niacin flush
Selenium 50 mcg	None	None	Metallic taste, garlic breath (rare)	Selenium toxicity (months of high dose only)	Blood test at 3 months.

A10.3 Phase 2 Active Treatment (Months 4-12)

Methylation Support:

Agent	Expected / Common	Management
Methyl B12 injections	Pain at injection site (5 min), mild redness	Rotate sites (arm, thigh, glute). Ice after injection. Use 30G needle for less pain
Methyl B12	Increased energy, hyperactivity in some	If hyperactivity occurs, switch to morning only. May be a positive sign (methylation waking up)
L-methylfolate	None typically. Can cause irritability in undermethylators	Start low (200 mcg), titrate up
TMG 500-1,000 mg	None. Loose stools at higher doses. Can cause hyperactivity	Give in morning
SAMe 200-400 mg	None. Can cause anxiety/agitation in some	Start low. Contraindicated in bipolar

Glutathione & NAD+:

Agent	Side Effects	Management
Liposomal glutathione	None typically — well tolerated	None
Nebulized glutathione	Coughing (first few sessions), throat irritation	Dilute with saline, start with shorter sessions (3 min), increase slowly
NAC 500 mg 2x/day	Nausea (rare at this dose), loose stools	Give with food. Reduce to 250 mg 2x if needed
Alpha-lipoic acid	Skin rash (rare), nausea (rare)	Give with food. ALA can lower blood sugar — monitor if diabetic
NR (nicotinamide riboside) 125 mg	None at this dose	None
Apigenin 50 mg 2x/day	None — natural flavonoid	None

Mitochondrial Support:

Agent	Side Effects	Management
CoQ10 (ubiquinol) 100-200 mg	None	Take with fatty meal for absorption
L-carnitine 500 mg 2x/day	Fishy body odor (20-30%), loose stools	R-ALA form reduces odor. Deodorant. Reduce dose if diarrhea
Creatine 2-3 g	None at this dose	Ensure adequate water intake
MCT oil 1-2 tbsp/day	Loose stools (see A10.1), abdominal cramping	Titrate slowly. Accept soft stools as adaptation

Anti-Excitotoxicity:

Agent	Side Effects	Management
Sulforaphane 10-50 mg	Mild gas, bloating (first week)	Start at 10 mg, increase by 10 mg weekly. Give with food
Taurine 250-500 mg 2x/day	None — very well tolerated	None
Memantine (off-label) 2.5-10 mg	Dizziness, headache, drowsiness (initial), constipation	Start 2.5 mg, increase slowly over 4 weeks. Prescriber must monitor

A10.4 Gentle Detox (Months 4-6)

Agent	Expected	Common	Occasional	Management
Chlorella 2 g 2x/day	Green stools	Gas, bloating (first week)	Nausea, green diarrhea	Start 500 mg, titrate up over 2-3 weeks. If severe GI, switch to broken-cell chlorella
Cilantro tincture	None	None	Mild increased urination, transient agitation (Hg mobilization)	If agitation occurs, ensure binder (chlorella) taken simultaneously to prevent redistribution
Silica water 1 L/day	None	None	None	Drink throughout day, not all at once
Malic acid 500 mg 2x/day	None	Sour taste	Mild GI upset	Give with meals, mix into food if taste objectionable
Infrared sauna	Sweating, thirst	Fatigue, mild headache (first sessions)	Dizziness, lightheadedness	Hydrate before/after. Start 10 min, increase 2 min/session. Cool-down period
Epsom salt baths	Relaxation, improved sleep	None	None	Ensure child does not drink bathwater. Rinse after

Herxheimer (detox) reaction: About 15-25% of children experience a temporary worsening of symptoms when detox begins — more stimming, irritability, loose stools, or fatigue. This typically lasts 3-7 days and is a positive sign (toxins mobilizing).

Management: Reduce dose by half, increase fluids, Epsom baths daily, and wait for it to pass before increasing dose again. **Do not stop treatment** — stopping and restarting causes repeated Herx reactions.

A10.5 Chelation-Specific Side Effects (Month 6+)

Agent	Expected	Common	Requires Monitoring	Emergency
DMSA	Mild increased urination, metallic taste	Loose stools, transient rash	Liver enzymes (ALT/AST) monthly, kidney function (BUN/Cr) monthly	Severe rash, anaphylaxis (rare), Stevens-Johnson (extremely rare) — stop immediately
ALA	None	Skin tingling (mild, passes)	Liver enzymes	Severe skin reaction (stop)
DMSA + ALA combined	None (expected synergy)	Temporary increase in stimming/irritability (days 2-4 = mobilization phase)	Same as individual	Same as individual

Brightening window: 60-70% of children show a temporary improvement in cognition, eye contact, and language on DMSA pulse days 2-3. This may be followed by a 1-2 day regression as redistribution occurs — this is expected and passes.

Mineral depletion monitoring: DMSA removes zinc, copper, and selenium along with heavy metals. This is why off-day replacement (A7) is critical. Signs of deficiency: zinc = white spots on nails, copper = fatigue/pale skin, selenium = muscle weakness. Blood test at 3-month mark.

Cholinergic Note (v1.4): See **Section 2.4A** for full discussion of nAChR involvement in ASD, how existing treatments support receptor recovery, and evidence-based nicotine options under specialist supervision. are implicated in ASD pathophysiology via *CHRNA7* gene variants (15q13.3), postmortem findings of reduced $\alpha 7$ nAChR binding, and heavy-metal cholinergic toxicity. Transdermal nicotine has shown preliminary efficacy for aggression/irritability in ASD (Lewis 2018). The methylation, glutathione, and chelation arms of this plan may support nAChR recovery. See [research/cholinergic_nicotine/](#) for full references.

Comprehensive Autism Treatment Plan — Suggested Biomedical Protocol v1.5

Based on 301 research papers | May 31, 2026 | Shareable Protocol Model

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